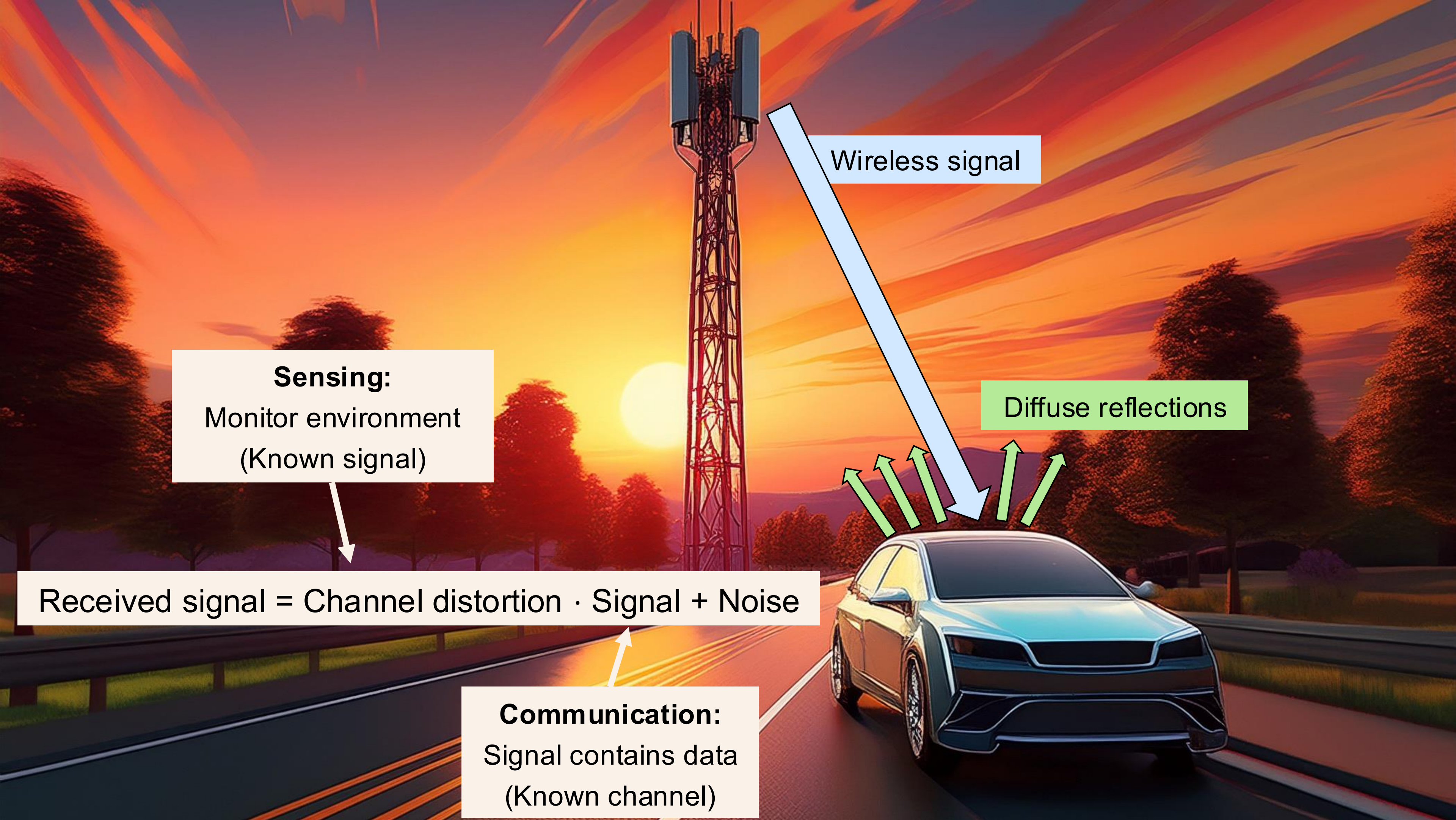
Introduction to **Integrated Sensing and Communication (ISAC)**

Emil Björnson

Professor of Wireless Communication

Fellow of IEEE, Digital Futures, and Wallenberg Academy

KTH Royal Institute of Technology, Stockholm, Sweden


Wireless signal
Diffuse reflections
Sensing:
Monitor environment
(Known signal)
Received signal = Channel distortion · Signal + Noise
Communication:
Signal contains data
(Known channel)

Traditional **Sensing** and **Communication** Systems

(Amada44, CC BY-SA 3.0)

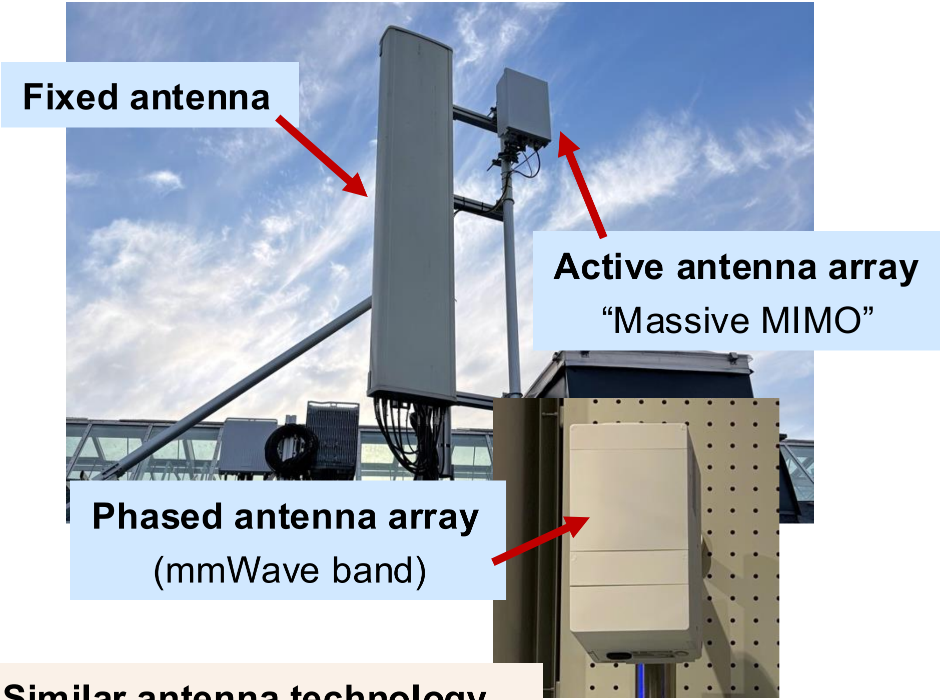
Marine 2D radar

(Spaningsradar 861)

Passive electronically scanned array (PESA)

(Lockheed Martin TPY-4)

Active electronically scanned array (AESA)

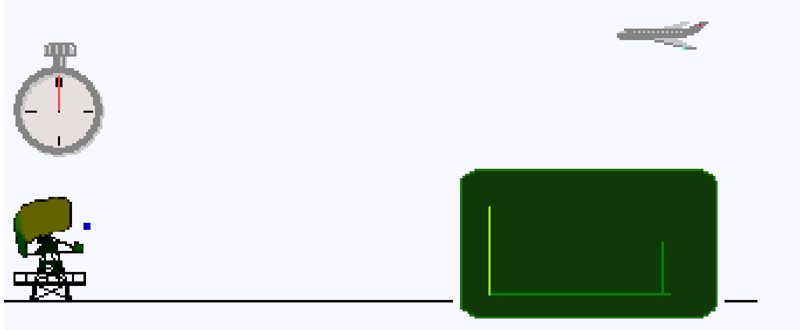


Similar antenna technology
Optimized for different use cases

Three Classical Radar Features

Radar = Radio Detection and Ranging

1) Measure delay → Estimate range

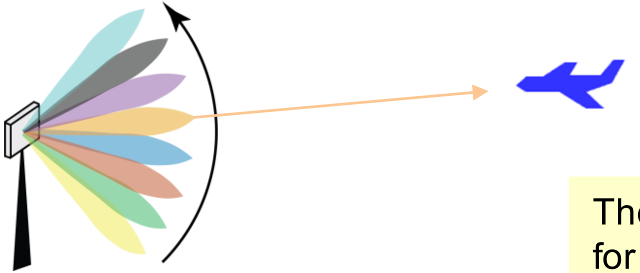


(Averse, Wikipedia, CC BY-SA)

2) Measure Doppler shift → Estimate velocity

(CassidyGE, Wikipedia, CC BY-SA)

3) Beam sweep → Estimate angle



These features can be used for detection and/or tracking

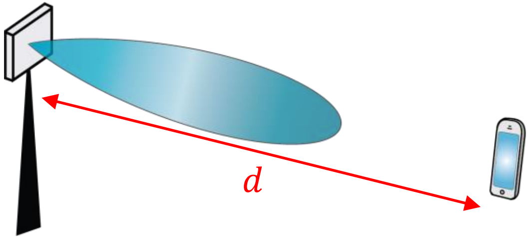
Similar Propagation Modeling

Radar equation:

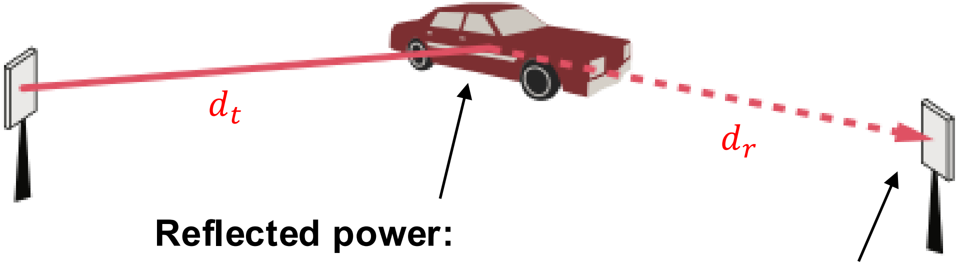
$$P_r = P_t \cdot G_t \cdot G_r \cdot \frac{\sigma_{\mathrm{RCS}}}{d_t^2 d_r^2} \frac{\lambda^2}{(4\pi)^3}$$

where $A_{\mathrm{r}} = G_r \frac{\lambda^2}{4\pi}$

Communication: Line-of-sight



Radar: Line-of-sight



Received power:

$$P_r = P_t \cdot G_t \cdot \frac{A_r}{4\pi d^2}$$

Area of receiver

Transmit power

Transmit antenna/array gain

Reflected power:

$$P_{\mathrm{reflect}} = P_t \cdot G_t \cdot \frac{\sigma_{\mathrm{RCS}}}{4\pi d_t^2}$$

Radar cross section (RCS) in m²

Received power:

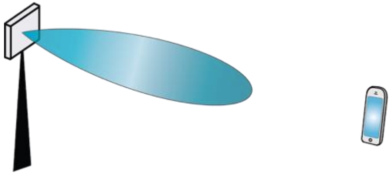
$$P_r = P_{\mathrm{reflect}} \cdot \frac{A_{\mathrm{r}}}{4\pi d_r^2}$$

Random RCS modeling

Average depends on size, shape, and material

- Swerling 1 (Rayleigh): σ_{RCS} has χ_2^2 distribution
- Swerling 3 (Rician): σ_{RCS} has χ_4^2 distribution

Important Differences



Downlink scenario

Uplink scenario

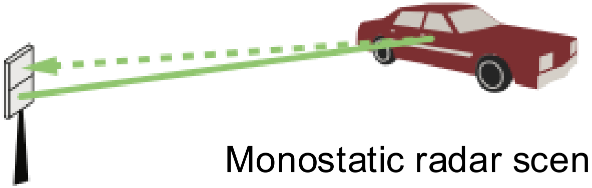
One-way link

Uplink/downlink separation

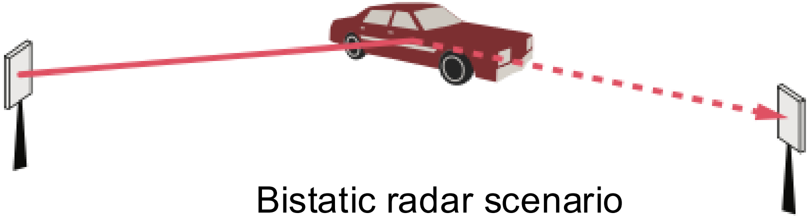
Half-duplex radios (in TDD)

Less propagation loss

3 GHz band



Monostatic radar scenario



Bistatic radar scenario

Two-way link

Only one mode of operation

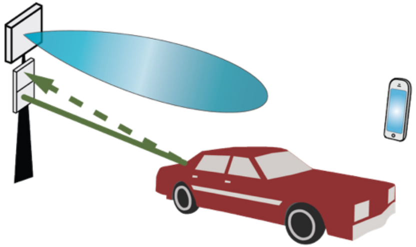
Full-duplex radios

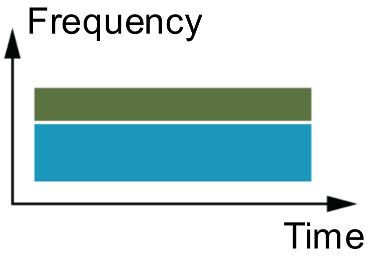
Much larger propagation loss

TDD = Time-division duplexing

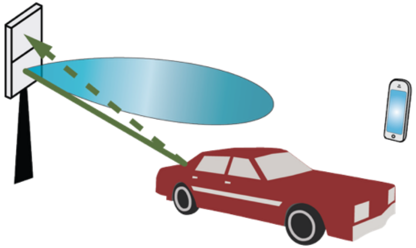
Three Levels of ISAC (Integrated Sensing and Communication)

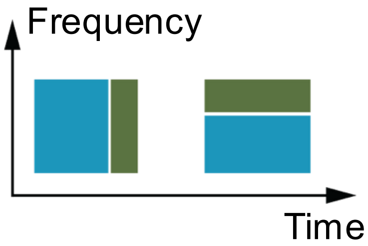
1) Site sharing



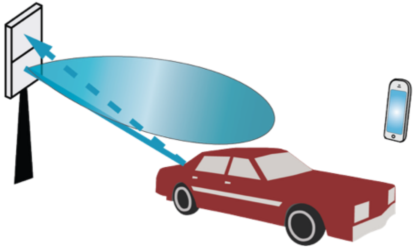


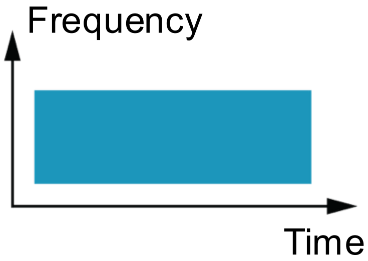
2) Hardware sharing





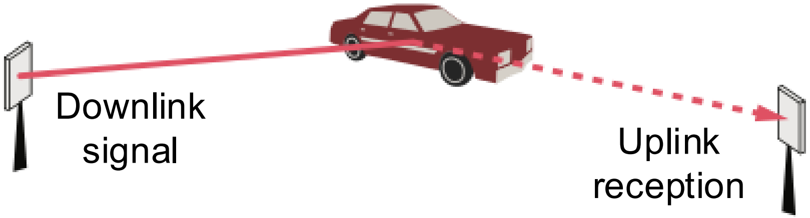
3) Signal sharing





Options for 6G?

My take on 6G ISAC



Bistatic sensing in a TDD band

- Primarily: communications
- Secondarily: sensing

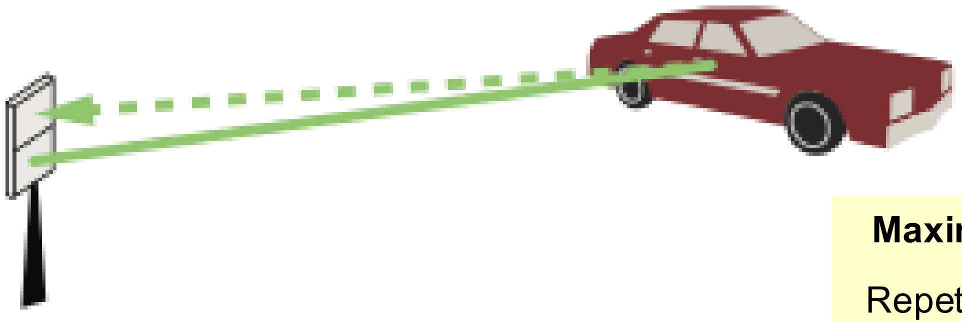
Network-internal use cases

- Track inactive users (Beamforming, handover)
- Adapt sleep cycles to activity and motion
- Enhanced inter-cell interference cancellation
- Enhanced positioning

Data collection for external applications

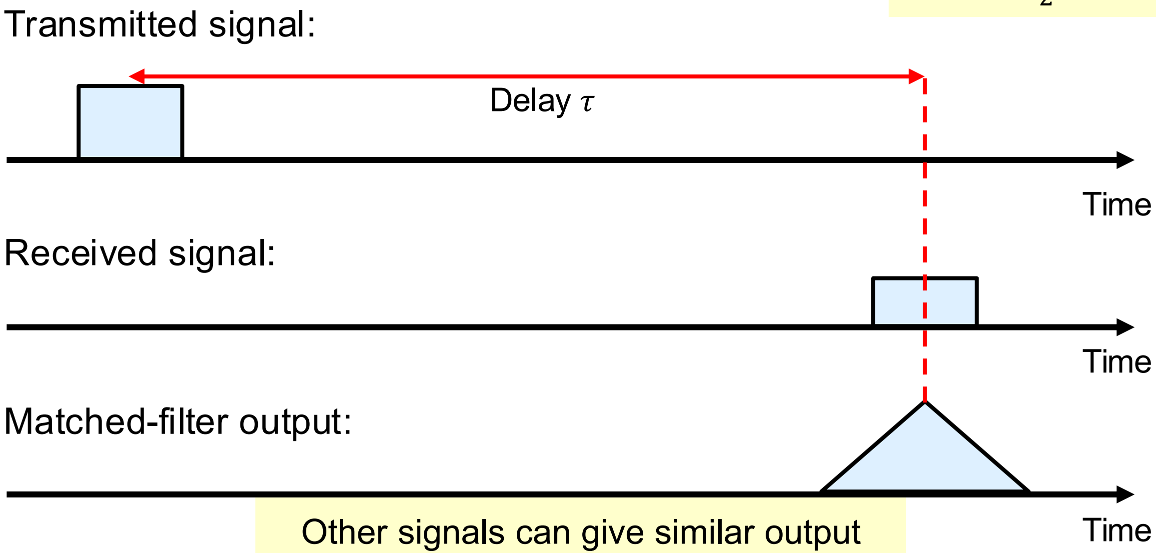
- Environmental awareness (human motion/gestures)
- Smart city optimization (monitor traffic and crowds)
- Smart factories (track robots, detect irregularities)
- Surveillance (drone detection, see in darkness)
- Data collection for digital twins and AI

Example: Range Estimation



Maximum range

$$\frac{\text{Repetition time}}{2} \cdot c$$



Range estimate:

$$d = \frac{\tau}{2} \cdot c$$

Speed of light

Range resolution

Bandwidth: B

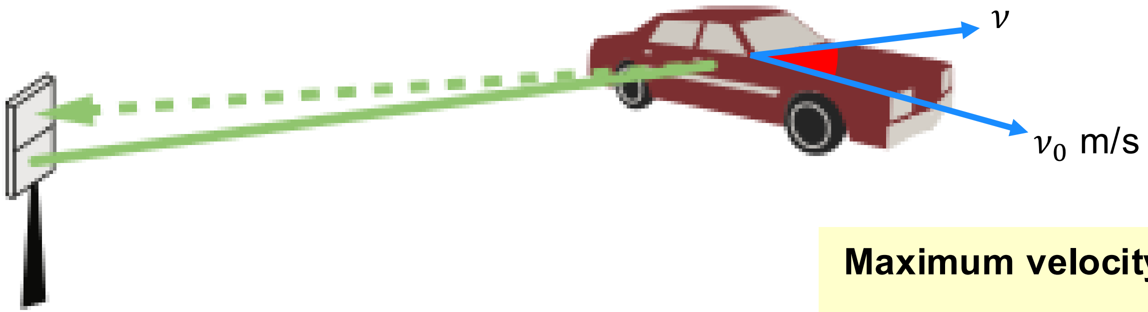
Pulse length: $\frac{1}{B}$

Time resolution: $\Delta\tau = \frac{1}{B}$

(Nyquist sampling)

Resolution on range: $\frac{\Delta\tau}{2} \cdot c = \frac{c}{2B}$

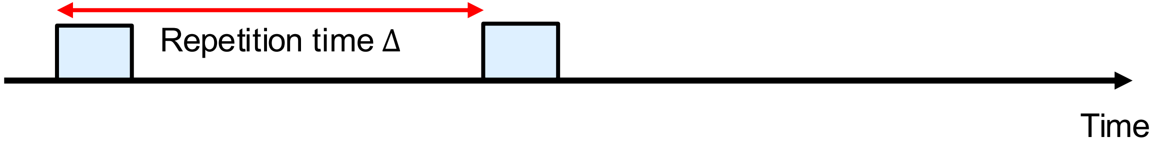
Example: Velocity Estimation



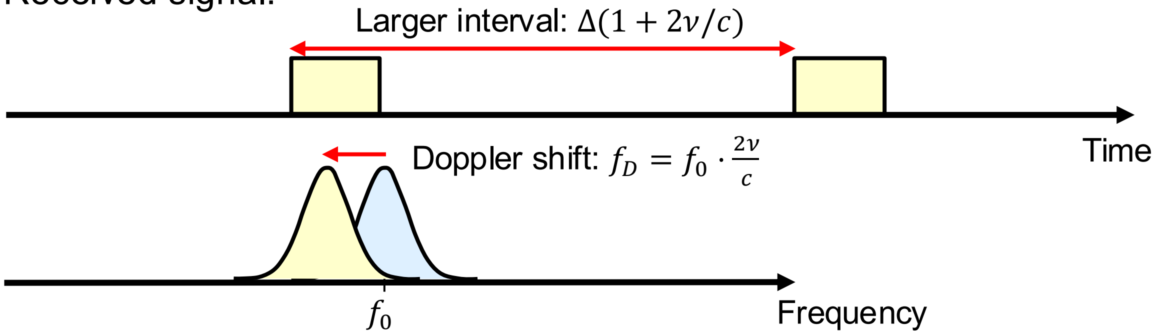
Maximum velocity

$$\frac{1}{2\Delta} \cdot \frac{\lambda}{2}$$

Transmitted signal:



Received signal:



Velocity estimate:

$$v = \frac{f_D}{f_0} \cdot \frac{c}{2} = \frac{f_D \lambda}{2}$$

Wavelength

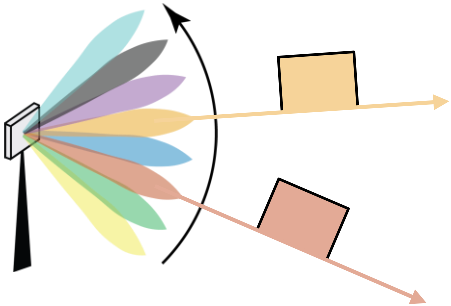
Velocity resolution

Total signal duration: T

Frequency resolution: $\Delta f = \frac{1}{T}$
(width of "Doppler bins")

Resolution on velocity: $\frac{\Delta f \cdot \lambda}{2} = \frac{\lambda}{2T}$

Example: Angle Estimation



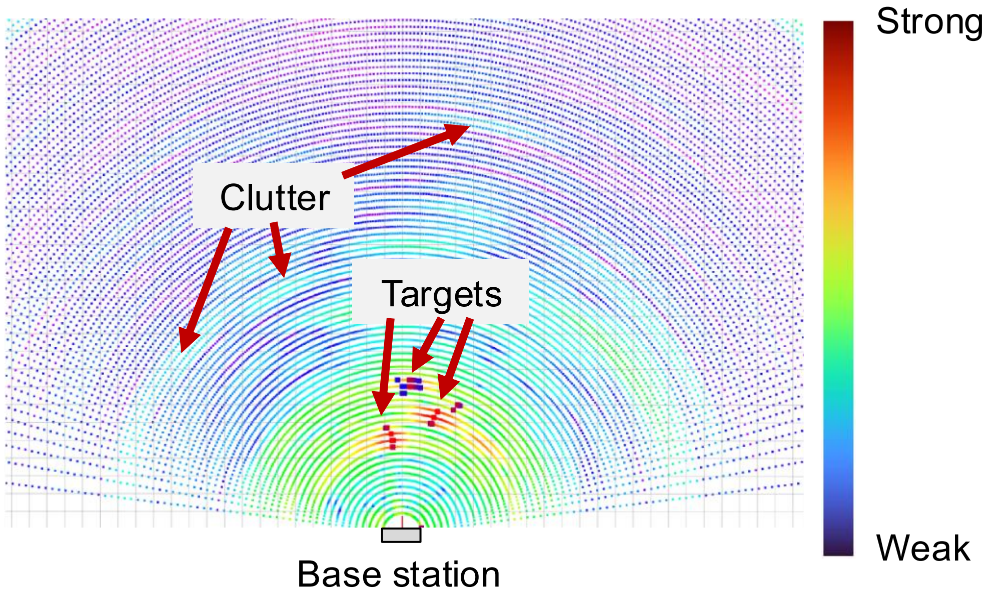
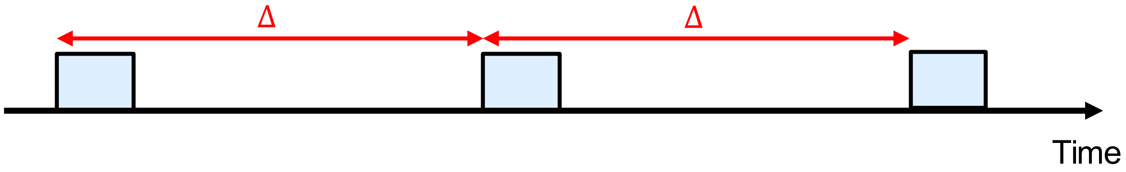


Image source: J. Steinbaeck, C. Steger, E. Brenner, G. Holweg, N. Druml, "Occupancy Grid Fusion of Low-Level Radar and Time-of-Flight Sensor Data," DSD 2019

Waveform Design: Range vs. Velocity Resolution



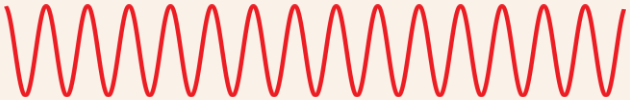
Opposite dependence on Δ:

Maximum range: $\Delta \cdot \frac{c}{2}$

Maximum velocity: $\frac{1}{\Delta} \cdot \frac{\lambda}{4}$

Example 1: Continuous-Wave (CW)

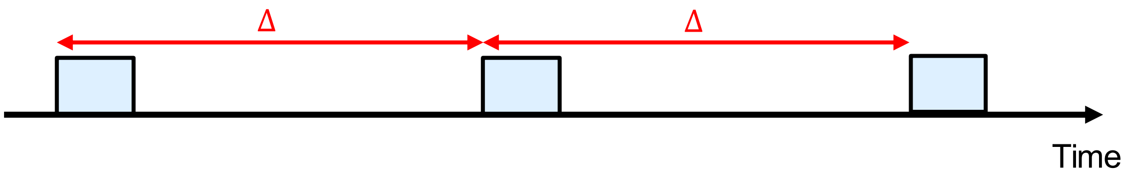
Transmit pure sinusoid:



Shifted received frequency:

Enables only velocity estimation

Waveform Design: Range vs. Velocity Resolution



Opposite dependence on Δ: **Maximum range:** $\Delta \cdot \frac{c}{2}$ **Maximum velocity:** $\frac{1}{\Delta} \cdot \frac{\lambda}{4}$

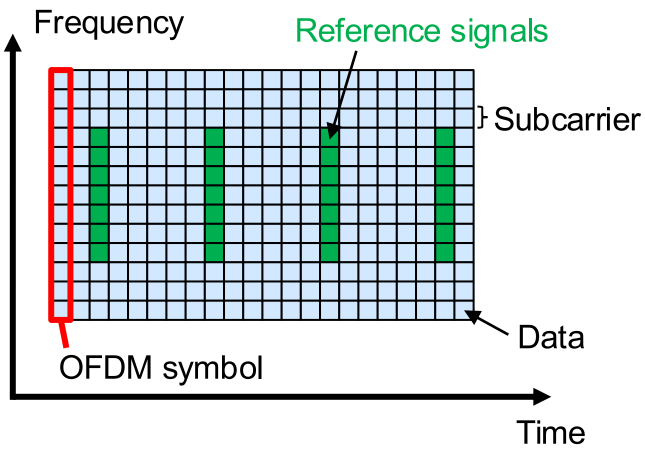
Example 2: Orthogonal-Frequency Division Multiplexing (OFDM)

Much bandwidth

Use reference signals or data (known at receiver) for radar sensing

Estimate the range within an OFDM symbol

Estimate the velocity OFDM across symbols



Summary

Integrated sensing and communication

- Add sensing functions to telecom systems
- Estimate range, velocity, and angle
- Internally: Enhanced resource allocation
- Externally: Input to advanced applications